

Nikhil Rout

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🐙 NikhilRout 🔗 NikhilRout.github.io

Education

Bachelor of Technology, Electronics and Communication Engineering 2022 – 2026
Vellore Institute of Technology, Chennai CGPA: 9.15/10

Research Experience

Visiting Research Assistant Los Angeles / Remote
Vortex GPGPU, ORCAS Lab, UCLA May 2025 – Present

- Developed a Tensor Core Unit (TCU) extension for the open-source RISC-V Vortex GPGPU, enabling WMMA/WGMMA-style acceleration for deep-learning kernels
- Architected a configurable Microscaling (MX) Fused Dot-Product datapath for prototyping custom low-precision formats and evaluating their E2E Perplexity–PPA tradeoffs on a programmable GPU
- Proposed LNSF4, a hierarchical logarithmic block-scaling format that achieves NVFP4-comparable model accuracy while reducing FEDP area and power consumption by $\sim 6\%$ and $\sim 11\%$, respectively
- Explored Tensor Core microarchitecture optimizations including format-shared multipliers, addend splitting for maximum accumulator reuse in fused INT–FP datapaths, and sparse-lane-mask clock gating for dynamic-power reduction in workloads exhibiting dual-sided unstructured sparsity
- Achieved 4-cycle FEDP latency at 300 MHz on the AMD Alveo U55C FPGA, delivering $\sim 2.7\text{--}7.9\times$ higher throughput over Berkeley HardFloat and FPnew at $< 60\%$ of their area cost, with end-to-end speedups on FlashAttention ($9.6\times$), Llama-3.2-1B ($16.7\times$), MoE ($10.3\times$), and NeRF ($6.8\times$)

Summer Research Intern Chennai
Centre for Nanoelectronics and VLSI Design, VIT Chennai May 2024 – Jul 2024

- Investigated optimal dataflow in radix-2 DIT and DIF butterfly structures for FFT accelerators
- Designed Number Theoretic Transform (NTT) cores with modulo-arithmetic operations for efficient polynomial multiplication in lattice-based post-quantum cryptography schemes on FPGAs

Research Publications

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- [1] **Nikhil Rout**, “RoadBlock: Rethinking GPU Tensor Core Microarchitecture for Emerging Microscaling Format Support”. In *ACM Parallel Architectures and Compilation Techniques (PACT)*, 2026.
 - [2] **Nikhil Rout**, Blaise Tine, “Ten-Four: An Open-Source Fused Dot Product Unit for Mixed-Precision SoftGPU Tensor Cores”. In *Preprint, Under Review*
 - [3] Yanggon Kim*, **Nikhil Rout***, Chengxuan Wang, Jacob Leigh, Xinle Song, Blaise Tine, “FourShadow: Investigating SIMT Tensor Core Scaling Across Hierarchies of a RISC-V GPU”. In *Under Review (Round 2)*
 - [4] **Nikhil Rout**, Joe Samuel Rosh, Blaise Tine, “SIMTurn: Lightweight Divergent Subgroup Scheduling for Intra-Warp Synchronization”. In *Under Review*
 - [5] **Nikhil Rout**, Jean Jenifer Nesam J, “Optimizing RGB to Grayscale, Gaussian Blur and Sobel-Filter operations on FPGAs for reduced dynamic power consumption”. In *IEEE Artificial Intelligence For Internet of Things (AIIoT)*, 2024.

Talks and Workshops

- [TinyVortex: Learning-Guided Kernel Scheduling for Configurable GPU Co-Design Space Exploration](#)
Workshop on Learning-Augmented Compilers & Systems (LACS) @ PACT 2026
- [Rethinking Tensor Core Microarchitecture for Emerging Microscaling Format Support on GPUs](#)
Workshop on Open-Source Computer Architecture Research (OSCAR) @ ISCA 2026
- [Exploring Fused Dot Product Microarchitecture for the Vortex Tensor Core Unit Extension](#)
Vortex Workshop and Tutorial (VORTEX) @ MICRO 2025
- Getting started with the Vortex GPGPU Framework and Tensor Core Unit Extension
Upcoming, RISC-V International Technical Sessions 2026

Skills and Competencies

Languages: Verilog/SystemVerilog, C/C++, OpenCL, CUDA, HIP, Python, PyTorch

Tools: Quartus Prime, Xilinx Vivado/Vitis HLS, Synopsys VCS/DC, Make, Git, LaTeX

Simulators: Vortex SimX, Accel-Sim/GPGPU-Sim, gem5, Ramulator

Honors, Awards and Volunteering

- Short-Term Research Scholar, UCLA CS (\$27K)
- Second Round (Ongoing), ACM SRC at PACT 2026
- Student Travel Grants - PACT'26 (\$400), ISCA'26 (\$1K), MICRO'25 (\$580)
- Best Paper Award, IEEE AIIoT'24
- Course Grader, UCLA (GPU Microarchitecture - Fall'26)
- Teaching Assistant, VIT Chennai (FPGA System Design - Spring'25)
- Volunteer, HotChips'26
- Community Volunteer, NSF POSE (RISC-V OpenGPU)

References

Dr. Blaise-Pascal Tine, *Assistant Professor, UCLA*

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Dr. Jean Jenifer Nesam J, *Assistant Professor, VIT Chennai*

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